

BENCHMARK: NON-DETECTION

Results

Mid-Transit Time (Tmid)	2458639.95588 +/- 0.00091 BJD_TDB
Ratio of Planet to Stellar Radius (Rp/R*)	0.001 +/- 0.045
Transit depth (Rp/Rs)^2	4.7e-05 +/- 0.0061 [%]
Orbital Inclination (inc)	87.32 +/- 1.8
Airmass coefficient 1 (a1)	2.61 +/- 0.19
Airmass coefficient 2 (a2)	-0.89 +/- 0.79
Scatter in the residuals of the lightcurve fit is	7.4 %
Best Comparison Star	None
Optimal Aperture	3.57
Optimal Annulus	8.04
Transit Duration (day)	0.0651 +/- 0.0028

Accuracy vs published ephemeris (ExoClock/NEA)

Rp/R■ fit vs published	0.001 ± 0.045 vs 0.13031 (deviation 2.59σ)
Duration fit vs published	0.0651 d vs 0.0954 d (deviation 9.75σ)
Mid-time O–C	-1.4 min
Depth SNR	0.0
Residual scatter	7.4 %

Light curve

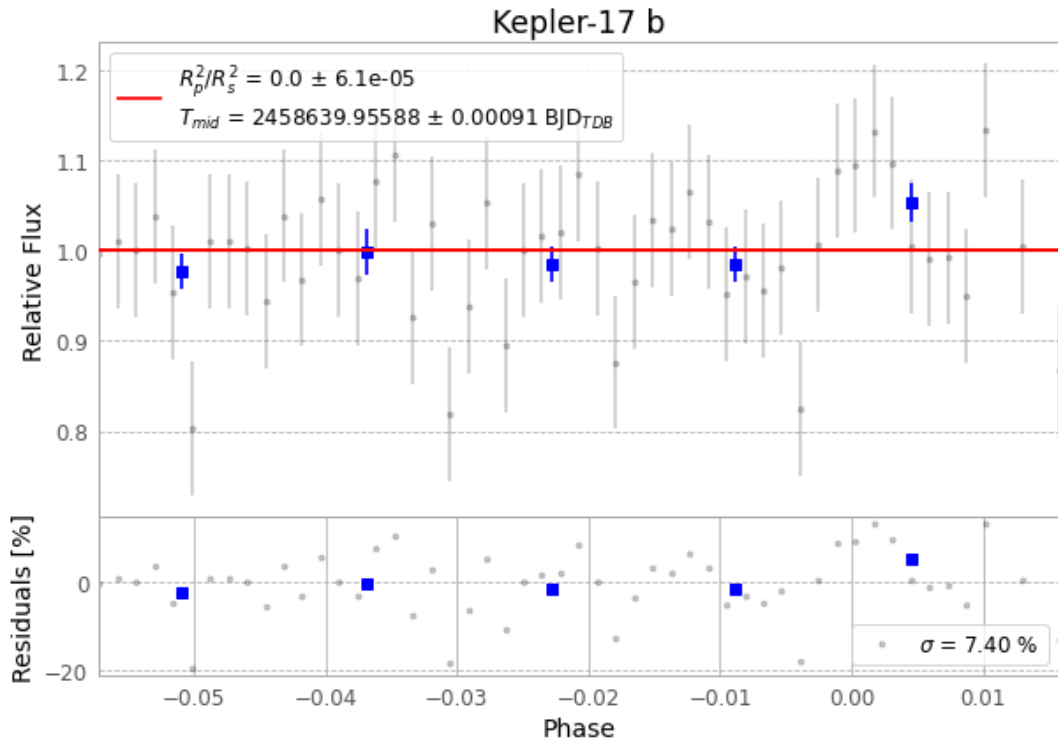


Figure 1 — FinalLightCurve_Kepler-17 b_2019-06-05.png (SHA-256 f4a3f45f5307ea13...)

Method & software

EXOTIC 4.3.1 aperture photometry with nested-sampling transit fit (ultranest 3.6.5); priors from the NASA Exoplanet Archive. Camera CCD, binning 1x1, filter CV, target pixel [423, 193], 1 comparison star(s). Exact configuration: parameters.inits_json in record.json (SHA-256 1d00ba268de28dd5...).

exotic 4.3.1 · astropy 6.1.7 · photutils 2.0.2 · numpy 1.26.4 · scipy 1.14.1 · twirl 0.5.1 · astroquery 0.4.11 · ultranest 3.6.5 · python 3.12.13 · pipeline_commit 1dc3d22

Data provenance

51 FITS frames (33 MB), KEPLER-17190605085112.FITS → KEPLER-17190605112716.FITS. Every input frame is SHA-256-fingerprinted in record.json; raw frames available on request and verifiable against that manifest. Artifacts: bench-KEPLER-17-190605.log (be4e91726d84...), FinalLightCurve_Kepler-17 b_2019-06-05.png (f4a3f45f5307...), FinalLightCurve_Kepler-17 b_2019-06-05.csv (16758fd6ae32...)

Compute resources

Wall time 71.5 s · peak memory None MB · host mac-mini-m1-8gb (macOS-26.4-arm64-arm-64bit)

Observer: — · AAVSO obscode ABGA · Automated pipeline with human review — nothing is submitted without a human approving this specific result. Data license CC-BY-4.0. Generated 2026-08-18 01:04Z.